

## Project 7 – Working with Python OS Module

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### Background

Learn and practice the basics of **Python OS** module.

### Objectives

1. Run simple Python programs by importing the OS module.
2. Document steps for inclusion in a project report to be submitted for review and grading in Blackboard.

### Prerequisites

1. A working **Linux VM** and internet connectivity.

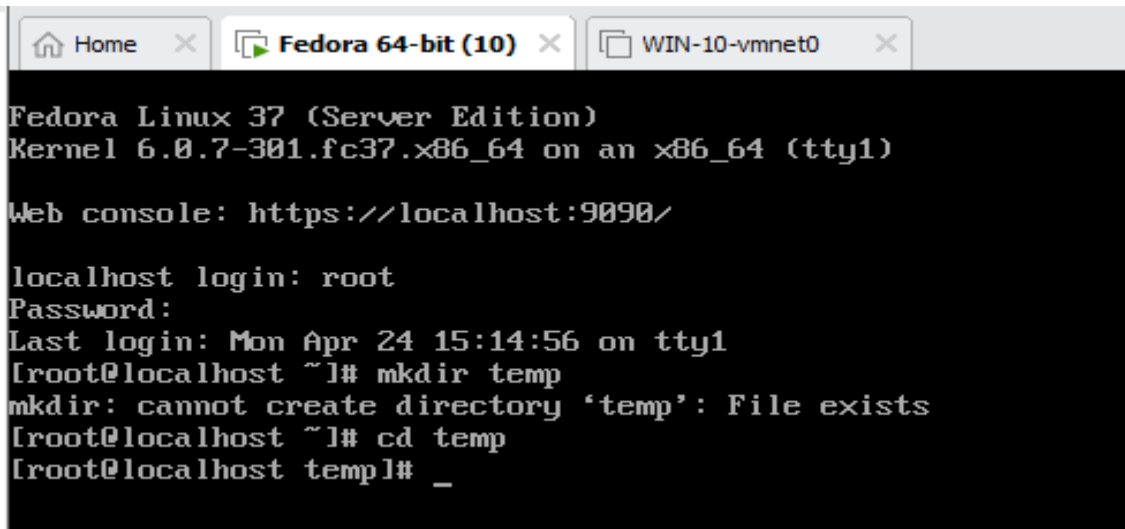
## Run a simple Python program using the OS Module

1. Start your **Linux VM** and login as the **root**.
2. Go to your home directory and **create a temp directory**.

```
# [mkdir temp]
```

3. Use command **cd temp** to enter the temp directory.

```
# [cd temp]
```



```
Home x Fedora 64-bit (10) x WIN-10-vmnet0 x
Fedora Linux 37 (Server Edition)
Kernel 6.0.7-301.fc37.x86_64 on an x86_64 (tty1)
Web console: https://localhost:9090/

localhost login: root
Password:
Last login: Mon Apr 24 15:14:56 on tty1
[root@localhost ~]# mkdir temp
mkdir: cannot create directory 'temp': File exists
[root@localhost ~]# cd temp
[root@localhost temp]# _
```



4. Create a `pythondir.py` file using the **vi editor** and input the following simple program that prints the working directory and creates a few sub-directories:

```
# [vi pythondir.py] : Create a file python use vi editor

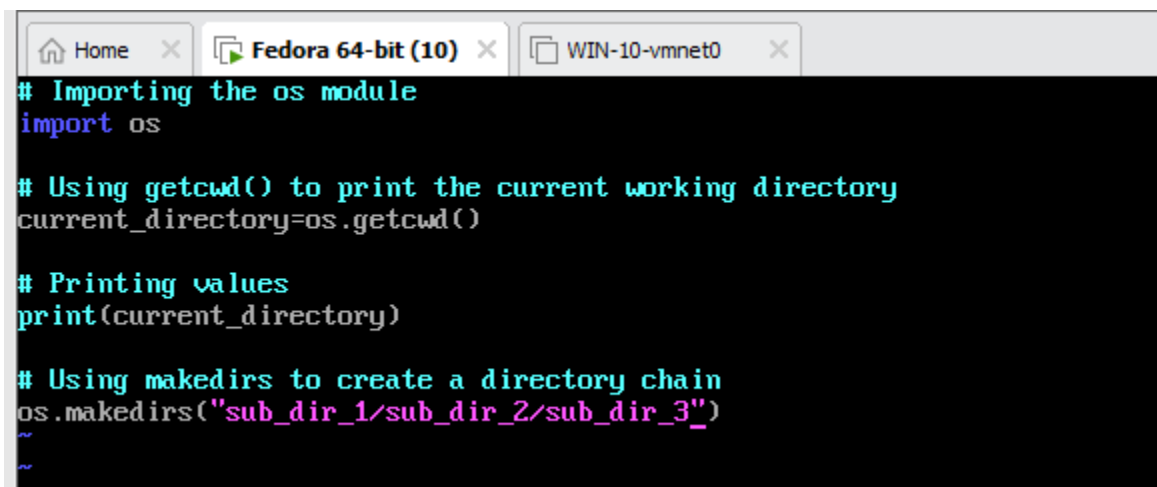
# importing the os module
import os

# Using getcwd() to print the current working directory
current_directory = os.getcwd()

# printing values
print(current_directory)

# Using makedirs to create a directory chain
os.makedirs("sub_dir_1/sub_dir_2/sub_dir_3")
```

5. Save and exit.



```
Home x Fedora 64-bit (10) x WIN-10-vmnet0 x
# Importing the os module
import os

# Using getcwd() to print the current working directory
current_directory=os.getcwd()

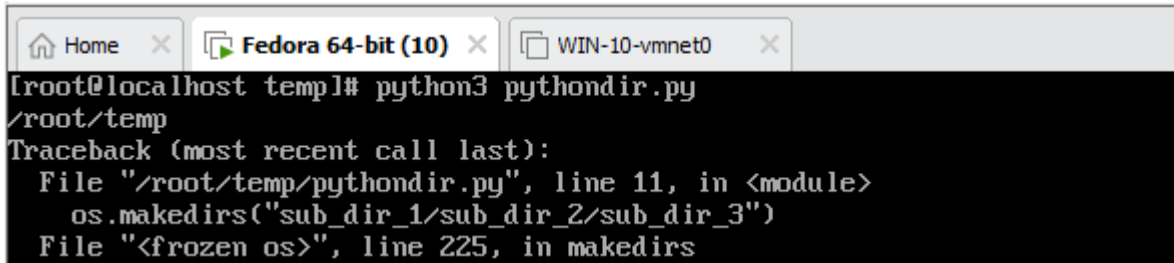
# Printing values
print(current_directory)

# Using makedirs to create a directory chain
os.makedirs("sub_dir_1/sub_dir_2/sub_dir_3")
~
~
```



6. Execute the program by typing `python3 pythondir.py`

# [`python3 pythondir.py`] : run file python



```
[root@localhost temp]# python3 pythondir.py
/root/temp
Traceback (most recent call last):
  File "/root/temp/pythondir.py", line 11, in <module>
    os.makedirs("sub_dir_1/sub_dir_2/sub_dir_3")
  File "<frozen os>", line 225, in makedirs
```

7. Check if those three subdirectories were created.

# [`cd sub_dir_1`] : go to directory sub\_dir\_1

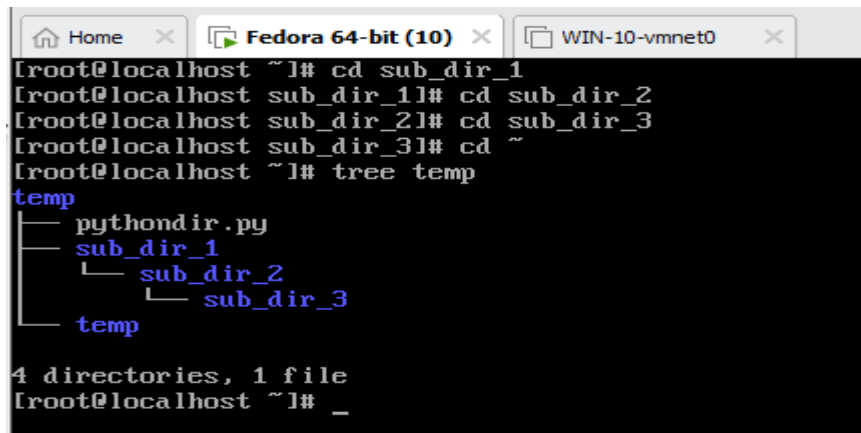
# [`cd sub_dir_2`] : go to directory sub\_dir\_2

# [`cd sub_dir_3`] : go to directory sub\_dir\_3

Or

# [`tree temp`] : Displays a directory tree containing the 3 newly created directories "`sub_dir_1/sub_dir_2/sub_dir_3`"

8. Include the output in your report.



```
[root@localhost ~]# cd sub_dir_1
[root@localhost sub_dir_1]# cd sub_dir_2
[root@localhost sub_dir_2]# cd sub_dir_3
[root@localhost sub_dir_3]# cd ~
[root@localhost ~]# tree temp
temp
├── pythondir.py
├── sub_dir_1
│   └── sub_dir_2
│       └── sub_dir_3
└── temp

4 directories, 1 file
[root@localhost ~]# _
```



## Additional commands in the OS Module

**Note:** for the following commands, demonstration is not required.

1. You can use the following command to remove the directory `os.rmdir('name of directory')`

```
# [os.rmdir temp] : Delete directory temp
```

2. To rename a directory `os.rename('directory's current name','new name')`

```
# [os.rename temp temp2] : change name directory temp is temp2
```

3. To change directory `os.chdir('path to the new directory')`

```
# [os.chdir temp/sub_dir_1/sub_dir_2/sub_dir_3]
```



## Ping a host using the OS Module

1. Create a `pythonping.py` file using the `vi editor` and input the following simple program. (Note your machine or virtual machine should be connected to the internet)

```
# [ vi pythonping.py ] : Create a file python use vi editor
```

(use `i` : insert)

```
# Importing the os module
```

```
import os
```

```
# Here we try to ping google.com
```

```
host = "google.com"
```

```
resp = os.system("ping -c 1 " + host)
```

```
# Checking the response
```

```
if resp == 0:
```

```
    print(host, 'is up!')
```

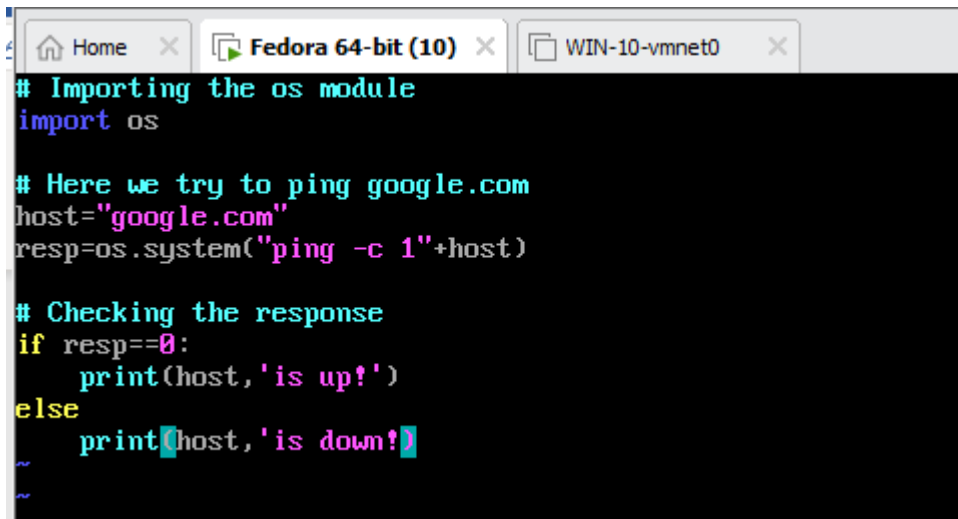
```
else:
```

```
    print(host, 'is down!')
```

2. `Save` and `exit`.

(use `esc` for quit; `:wq` for save and exit)





```
# Importing the os module
import os

# Here we try to ping google.com
host="google.com"
resp=os.system("ping -c 1"+host)

# Checking the response
if resp==0:
    print(host,'is up!')
else:
    print(host,'is down!')
```

3. Run the code using `python3` command.

```
# [python3 pythonping.py] : run file python
```

4. Attach your output here.

### Submission

Use this document (or one of your choosing) to submit your screen shots and any other relevant data. At a minimum, you should show the commands run after logging in as root. Please ensure your submission includes your name, class, date, and reference to this lab.

